

AdaptLBE: Toward Co-optimizing Real-Object Placement and Redirected Walking to Host an Existing VR World in an Ordinary Room

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ABSTRACT

Location-based entertainment (LBE) allows participants to experience VR while walking through a physical space. However, it requires a large dedicated tracking space and permanent equipment, which limits its use to specialized areas. One solution is to reuse the existing furniture as passive-haptic proxies while walking a larger VR world within a confined space. Previous studies address these needs separately, adapting either the virtual or the physical environment, or reusing single proxies via redirected walking (RDW). However, none of these studies designs proxy placement and RDW jointly across a scene. We propose AdaptLBE, which combines the two: from a single room scan, it compresses each piece of furniture toward the anchor centroid and builds an RDW gain map that absorbs the residual within perceptual thresholds, giving a VR area larger than the room without resets. We implement the proof-of-concept on one iPhone and one standalone Quest 3, with no PC or dedicated facility. In a pilot test (one participant, 20 seatings), the avatar walked a 23.3 m² VR area inside a 15.5 m² physical room—a 1.5× larger walkable area (expansion ratio $R = 1.50$)—with all gains within thresholds and a median chair-alignment error of 0.37 m.

Index Terms: Location-based entertainment, redirected walking, passive haptics, mixed reality, social metaverse.

1 INTRODUCTION

Location-based entertainment (LBE) lets participants walk on foot through a room-filling VR world at a public venue—an experience beyond consumer room-scale VR, which confines a single user to the few square meters of cleared floor tracked by a headset. Real walking is more immersive than controller or walking-in-place locomotion [1] and reduces cybersickness by limiting the visual-vestibular mismatch [2]. Today, however, this walk-through scale is reached only in dedicated LBE facilities with a large tracking space and permanent installations, whose cost has kept VR out of everyday spaces such as classrooms and meeting rooms.

For example, to gather students in a town-scale VR world for an avatar discussion, the usable floor is only about 15 m² of single-plane space once desks are pushed aside, no permanent equipment can be left after class, and the furniture that remains is scattered.

Because most VR worlds are designed without regard to physical correspondence, feasible approaches are limited. One adjusts the layout of either side: Substitutional Reality [3] and Scenograph [4] adjust the VR side to match the physical space, while TurkDeck [5] and RoomShift [6] reconfigure the physical side but need many hands or dedicated robots. The other is Redirected Walking (RDW), which walks a larger VR space within a limited physical one by



Figure 1: AdaptLBE turns an ordinary room into a location-based VR venue. (a) Chairs of an unmodified existing VR world that the avatar visits; (b) the same chairs re-placed in the physical room, compressed toward the anchor centroid.

accumulating virtual-physical offsets within perceptual detection thresholds [7, 8, 13]. Prior work, however, maximizes the coverage of individual *physical proxies* [9, 10, 11, 12] or transfers a physical layout to a virtual one [14], but none designs furniture placement and RDW *jointly* across a whole scene. Real furniture provides passive haptics as a proxy but is fixed, so its correspondence breaks down as RDW accumulates offset. Passive haptics and RDW expansion thus must be designed together.

We present AdaptLBE, the framework to couple the two, jointly designing where to place each piece of furniture and how RDW binds it to VR. The goal is to enable an ordinary room to reproduce an existing VR world without resetting, while still providing passive haptics.

2 SYSTEM AND METHOD

AdaptLBE uses a single room scan to design a placement target and an RDW gain map for each piece of furniture. The furniture is compressed toward the anchor centroid, and the residual is absorbed with within-threshold gains.

AdaptLBE runs on the Cluster metaverse platform as a one-to-one iPhone–Quest 3 pair. The iPhone (LiDAR, iOS 16+) handles scanning with Apple’s RoomPlan (a room-scanning API) and planning. Over Wi-Fi the pair exchanges Open Sound Control (OSC) messages: the Quest 3 streams the avatar’s pose, and the iPhone returns RDW gains that the Quest 3’s Cluster Creator Kit script applies. Each furniture hard-codes its metadata in a dedicated scene. The VR-physical bind sets translation to the anchor XZ centroid (origin O), orientation from a short alignment walk, and floor height from ARKit’s .floor plane, resolving RoomPlan’s indeterminate per-scan yaw by wall-plane matching.

A session has four stages. **Scanning and planning (S1–S2).** RoomPlan gives category-labeled 3D boxes of walls, floor, and furniture; from origin O the iPhone computes a compressed placement p_i (S3) and a per-segment gain map (S4).

Compressed placement of real furniture (S3). A furniture i at VR position v_i is placed at $p_i = O + (v_i - O)/\rho$, shrinking its displacement from the origin O by a uniform factor $1/\rho$ (compression ratio $\rho > 1$). Its physical distance from O is then $D_P = \|v_i - O\|/\rho$, its virtual distance from O shrunk by $1/\rho$. With $\rho > 1$, each piece is drawn inward, folding the physical layout toward the center.

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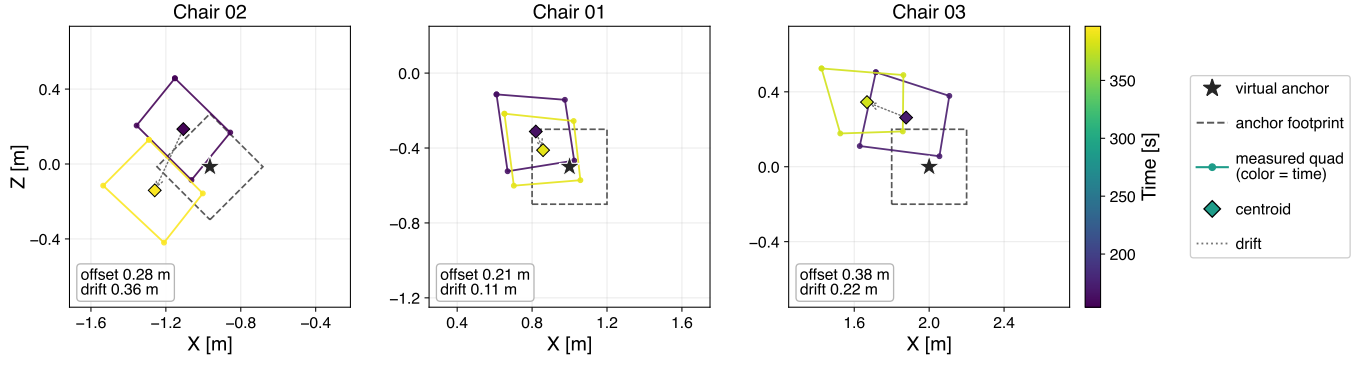


Figure 2: Virtual-real chair deviation, per chair (four-corner touch, one participant). Each panel shows the virtual anchor (*, dashed footprint) against the real chair measured at different times: the measured quads and their centroids ($\diamond$) are colored by time and linked by drift arrows (Cluster XZ frame).

Given the walkable area, the anchor set, and Steinicke et al.’s detection thresholds [7], a solver maximizes reachability to every proxy, then minimizes furniture movement. Since larger ρ reaches farther, the search is driven toward stronger compression until it hits the threshold-imposed upper bound; here it reached $\rho = 1.26$. ρ is fixed per session, unlike the time-varying gain g_T .

Gain-budget-based runtime RDW (S4). At runtime, RDW absorbs the residual left by this compression. Realizing a virtual segment over a physical one keeps their length ratio within Steinicke’s band $[0.86, 1.26]$; the shortfall is covered by a curvature gain g_C up to $\pm g_C^{\max} D_p$. We re-implement Steering—building on artificial potential field (APF) [15] and alignment [16] controllers—as Hirt et al.’s predictive APF [17] with clothoid prediction. The translation gain g_T and curvature gain g_C vary per frame with chair reachability (rotation $g_R = 1.0$ fixed); g_T stays within the band and drops below 1 near a chair to actively repay accumulated offset, reconciling proxy alignment with RDW expansion. Since S1–S3 run once per venue, participants start experiencing right after wearing the headset.

3 PRELIMINARY EVALUATION

We verified on real hardware whether the framework achieves larger-than-physical VR walking ($R > 1$) and chair alignment at seating under within-threshold gains without resets. One participant sat on three compression-placed chairs in a single session.

Metrics. The **expansion ratio** $R = A_{\text{virt}}/A_{\text{phys}}$ is the ratio of the XZ convex-hull areas the avatar and the body traverse ($R > 1$: a larger-than-physical VR walk). The **alignment error** e_{align} is the body–real-chair XZ offset when the avatar overlaps the chair (0: aligned passive haptics).

Condition. The physical space was a single-plane room (about 8.6×8.5 m, 45.5 m² walkable) scanned by LiDAR. Three chairs fixed at 1.0–2.0 m from the center in VR were compressed with $\rho = 1.26$ (Fig. 1), and one participant visited and sat on them through the avatar. We analyzed the 406 s after targeting began (about 16 fps, 20 seatings, RDW enabled), excluding the pre-start calibration and OSC-gap outliers; each chair’s real pose was recovered by four-corner touch.

(1) Expansion ratio. The VR-trajectory convex-hull area is $A_{\text{virt}} = 23.3$ m² against the physical $A_{\text{phys}} = 15.5$ m², giving $R = 1.50$ (physical walking 124 m, VR 157 m).

(2) Approach and convergence. Before each seating, the body-to-nearest-chair distance converged below 0.5 m, and 17 of 20 real seatings succeeded. Although RDW was enabled throughout, the visual–physical offset remained within a median of 0.48 m and a maximum of 1.45 m, so it does not diverge even without resets. All gains stayed within thresholds: translation $g_T \in [0.86, 1.26]$, $g_T < 1$ near chairs, curvature $\leq 4.94^\circ/\text{m} < 7.5^\circ/\text{m}$, and rotation $g_R = 1.0$.

(3) Alignment error. We took each chair’s physical seat as the planned placement point $p_i = O + (v_i - O)/\rho$ (S3, $\rho = 1.26$) and computed the distance to the physical body at each seating as e_{align} over the 20 trials. The error was a median of 0.37 m and a maximum of 0.60 m, with 19/20 below 0.55 m, comparable to a chair footprint (about 0.5 m). Thus the virtual–physical offset that would accumulate to several meters under uncorrected RDW was held to about one proxy width, and the seated-hip landings cluster near each planned seat. However, e_{align} does not separate success from failure (low-error failures occur), so it is not a predictor of success; the four-corner chair center sits 0.29 m from its virtual anchor on average (the *offset*; range 0.21–0.38 m) and shifts 0.23 m between the first and last measurement (the *drift*; range 0.11–0.36 m, Fig. 2)—a placement/registration residual that likely causes part of the error.

Under reset-free RDW the ratio is strongly constrained by thresholds (translation-gain area expansion is bounded by $g_T^2 \approx 1.6$). Thus the obtained $R = 1.50$, from a single translation compression and mild steering, is a realistic operating point at which the system keeps passive haptics.

4 CONCLUSION AND FUTURE WORK

We proposed AdaptLBE, which reuses at-hand furniture as passive-haptic proxies to reproduce an existing VR world in an ordinary room, coordinating furniture placement and RDW without resets. Running the full pipeline for one participant in one room, the iPhone–Quest 3 implementation yielded $R = 1.50$, within-threshold gains, and a chair-footprint-level alignment error—preliminary evidence that spatial expansion and furniture correspondence can be achieved together in an ordinary room.

Limitations and Future Work. This $n = 1$ study claims no significance: the measured R reflects this session’s path shape, not an upper bound on expansion, and seating did not always succeed, leaving room to improve alignment (e.g., a higher curvature limit). Repeated measurements with control conditions (RDW off / waypoint-only) will allow us compare R , e_{align} , and the haptic-constraint cost ΔR (the walkable-area expansion sacrificed to keep every proxy touchable) within subjects, and study how they depend on room size and shape. The central next step is a co-optimization solver that jointly solves placement and variable RDW with ΔR as the objective.

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